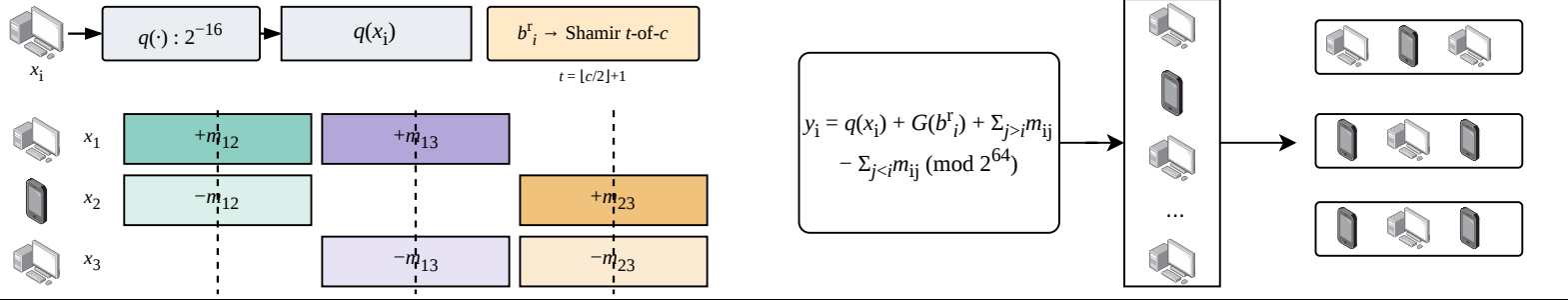


(b) Client masking



Legends

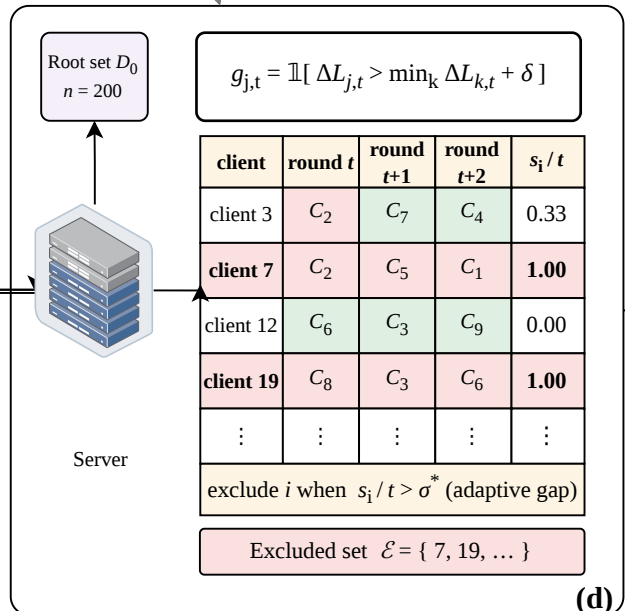
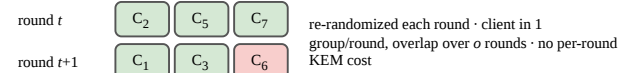
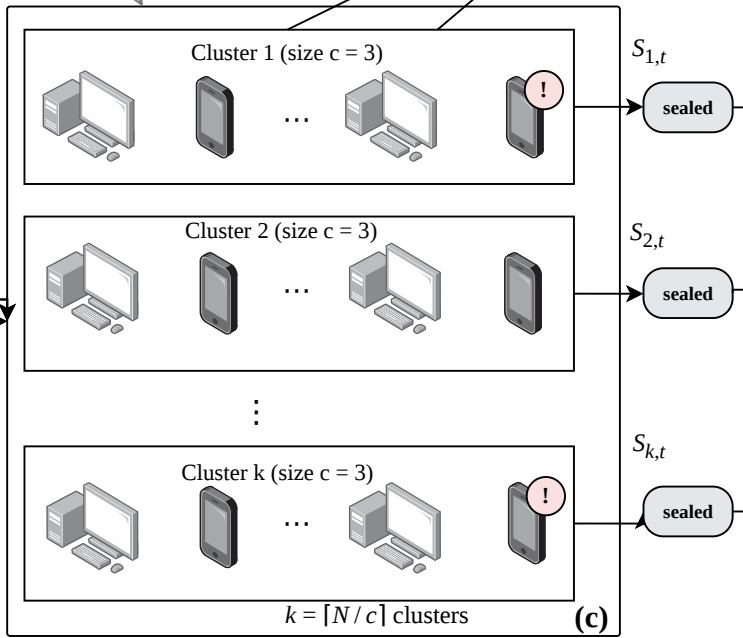
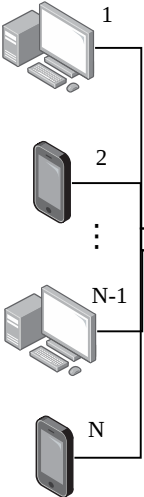
- malicious client
- clean / kept
- flagged / excluded
- sealed masked cluster sum $\rightarrow S_{j,t}$
- privacy boundary (server sees only aggregates)
- client's group in a round (re-randomized each round)

Clients

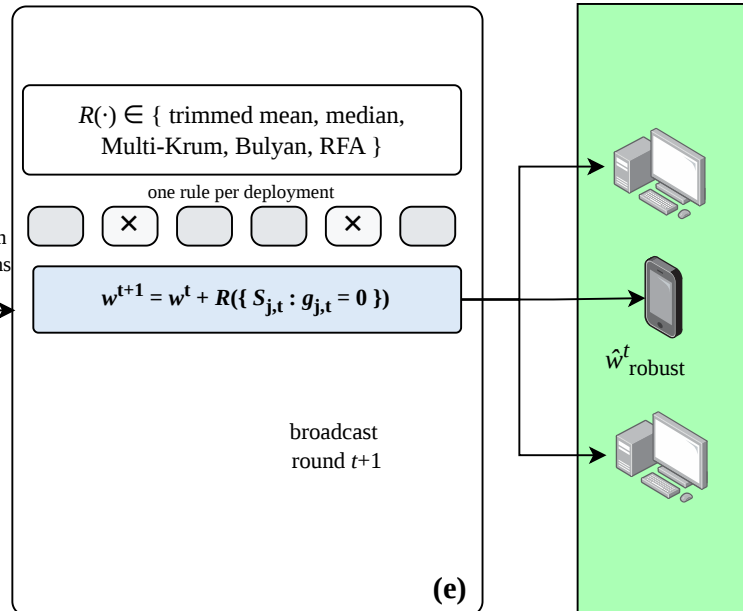
dropout-tolerant: any $c-t$ may drop

survivor: t shares

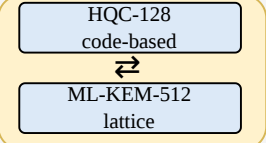
$\times$ dropout: recover via $H(ss)$



clean means $\hat{w}^t$



$K = (\text{KeyGen, Encaps, Decaps})$



H : HKDF-SHA256 · G : ChaCha20 KEMs : HQC / ML-KEM · 15.2 / 3.8 KiB, once

